

10.1 The gradient of a tangent as a limit.

$m = \frac{\Delta y}{\Delta x}$ → can be found on a curved line using a tangent. Then consider what will happen as the coordinates used to find the gradient get closer to the point of contact.

10.2: Differentiation using standard results.

$y = kx^n \Rightarrow \frac{dy}{dx} = knx^{n-1}$

$y = kx \Rightarrow \frac{dy}{dx} = k.$

$y = C \Rightarrow \frac{dy}{dx} = 0$

10.3 Tangents & normals

A tangent touches a point on a curve so it has the same gradient. A normal is a line that is perpendicular to a curve at the point of contact

$m_T = f'(n), \quad m_N = \frac{-1}{m_T}$ } use $y - y_1 = m(x - x_1)$

10.4 Increasing & Decreasing functions:

$\frac{dy}{dx} > 0 \Rightarrow$ increasing, $\frac{dy}{dx} < 0 \Rightarrow$ decreasing

$\frac{dy}{dx} = 0 \Rightarrow$ turning point ($m = 0$).

Nature of a TP can be seen by the gradients adjacent to the point.

10.5 Sketching x vs $\frac{dy}{dx}$ functions.

$\frac{dy}{dx}$ will be one order less than the original polynomial.

⇒ look for when $\frac{dy}{dx} = 0$

⇒ look for increasing & decreasing sections

10.6: Extending the rule → just differentiate bro

10.6: Higher Order Derivatives (getting abstract!)

Rate of change of gradient = $\frac{d}{dx}\left(\frac{dy}{dx}\right) / \frac{d^2y}{dx^2} / f''(x)$

To find the nature of a turning point, sub your value into $f''(x)$.

$\Rightarrow f''(n) > 0 \Rightarrow$ minimum

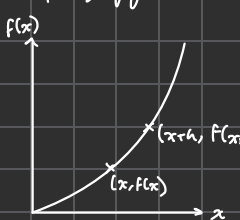
$\Rightarrow f''(n) < 0 \Rightarrow$ maximum

$\Rightarrow f''(n) = 0 \Rightarrow$ look at $\frac{dy}{dx}$ either side of TP, may be a point of inflection.

10.8: Practical Uses:

Differentiation can be used to find minima & maxima values of quantities based upon their base equations

10.9: Differentiation from first Principles:



$h = \delta x / \Delta x$

gradient of chord $= \frac{f(x+h) - f(x)}{x+h-x} = \frac{f(x+h) - f(x)}{h}$

To find the gradient at point x, $\delta x \to 0 \Rightarrow h \to 0$

$\Rightarrow m(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

Power Rule Derivation?

$f(x) = x^n$.

$\Rightarrow m_1 = \frac{(x+h)^n - x^n}{h} \quad (\lim_{h \to 0})$

$= \lim_{h \to 0} \frac{x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 \ldots - x^n}{h}$

$= \lim_{h \to 0} nx^{n-1} + \binom{n}{2}x^{n-2}h^2 \ldots$

$= nx^{n-1} = f'(x)$